Quantitative Action Quality Assessment

Parting the Wild Horse's Mane: form-level sequences

Segmented SMPL Sequences

One reference; a student cohort

Reference: QxVvRcRn2TA



Start ... End

Student: 10



Start ... End

Real tracking meshes
Canonical front view for display

(a) Uniform-Anchor Pose Comparison

Uniform Sampling

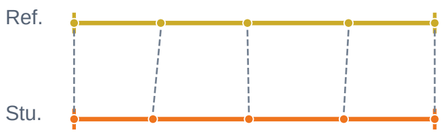
Reference anchors at 1 FPS

Include first and last frames

K fixed reference anchors

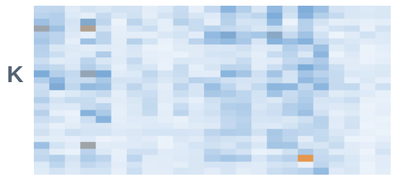
SMPL-DTW Matching

Local cost: mean SO(3) distance



For each anchor: minimum local-cost student candidate on the DTW path

Matched-anchor joint distances



23 local joints

Mean Pose Distance

Average over K anchors and 23 local joints; global orientation excluded.

(b) Pelvis Motion Comparison

Body-Relative Signals

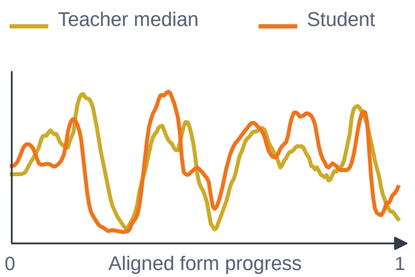
Pelvis relative to ankle midpoint

Height, lateral and forward motion

Normalize by body size

DTW-align to form progress

Lateral pelvis trajectory



Pelvis Deviation

Horizontal trajectory RMSE
Horizontal endpoint offset
Vertical trajectory RMSE

Teacher-calibrated excess

Fixed Weighted Sum

Pelvis reference: median of 10 teachers; robust leave-one-teacher-out calibration.

Equal-Weight Borda Fusion

Rank students within the same form

Pose rank

Smaller mean distance is better

Pelvis rank

Smaller deviation is better

50% pose + 50% pelvis

$$Q = 0.5q_{\text{pose}} + 0.5q_{\text{pelvis}}$$

$$q = (N - r)/(N - 1)$$

Relative Quality Ranking

Order by fused rank quality Q

Average ranks for ties; N > 1

Uniform sampling selects anchors only; DTW operates on the full sampled pose sequences.